

FinQuery

A local-first personal-finance analyst. Drop your bank statements into a chat, ask in your own words, get numbers you can audit.

 FinQuery

+ New chat

Dashboard

Transactions

Imports

Memory

Recent

The sample year

The sample year +

Dashboard

Every figure here comes from a query this page just ran. Ask for a chart in a chat; the ones worth keeping land here.

This monthLast 3 monthsThis yearAll - 01.01.2025 to 28.12.2025

Any date

Spent

2.346,34 €

December 2025

-73,80 € (-3 %) vs last month

-72,69 € (-3 %) vs 6-month average

Income

2.850,00 €

December 2025

0,00 € (0 %) vs last month

0,00 € (0 %) vs 6-month average

Net

503,66 €

December 2025

+73,80 € (+17,2 %) vs last month

+72,69 € (+16,9 %) vs 6-month average

Needs review

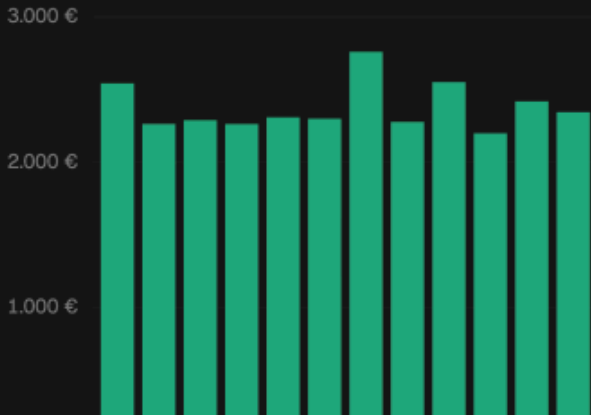
25

01.01.2025 to 28.12.2025

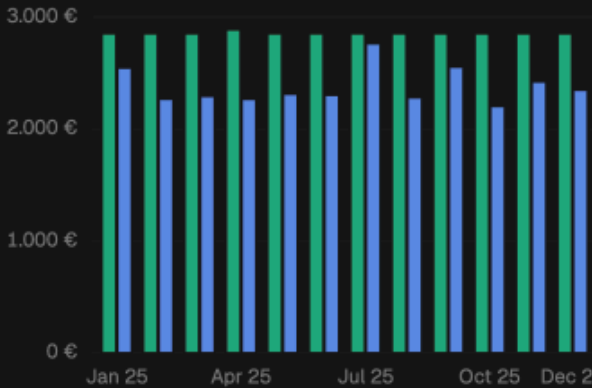
Bookings without a category

Answer them in a chat

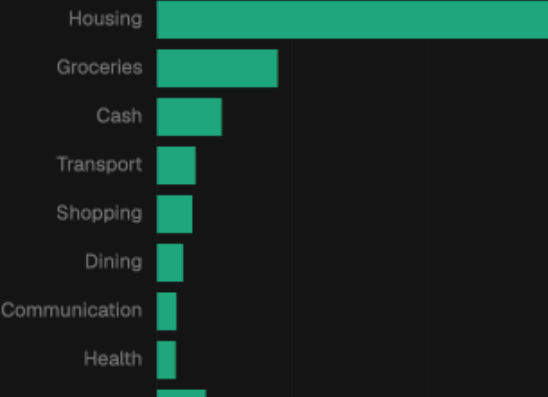
Spending per month



Income against spending per month



Top categories

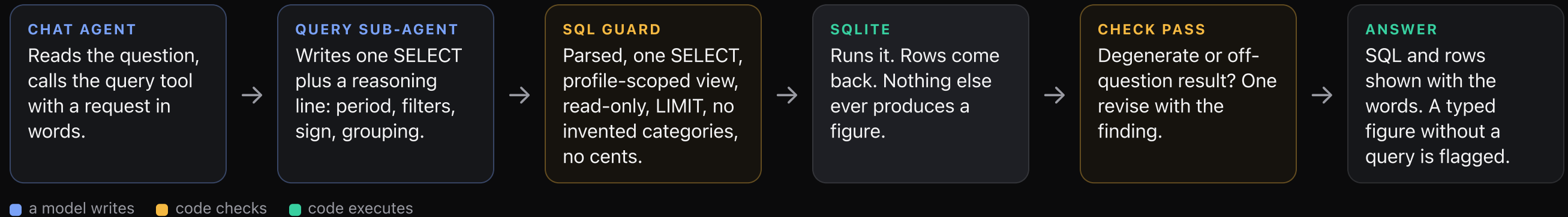


Rahul Singh and Liudvikas Zekas, Hasso Plattner Institute, Potsdam

Runs on the laptop: two Gemma 4 models through llama.cpp, SQLite, a React app

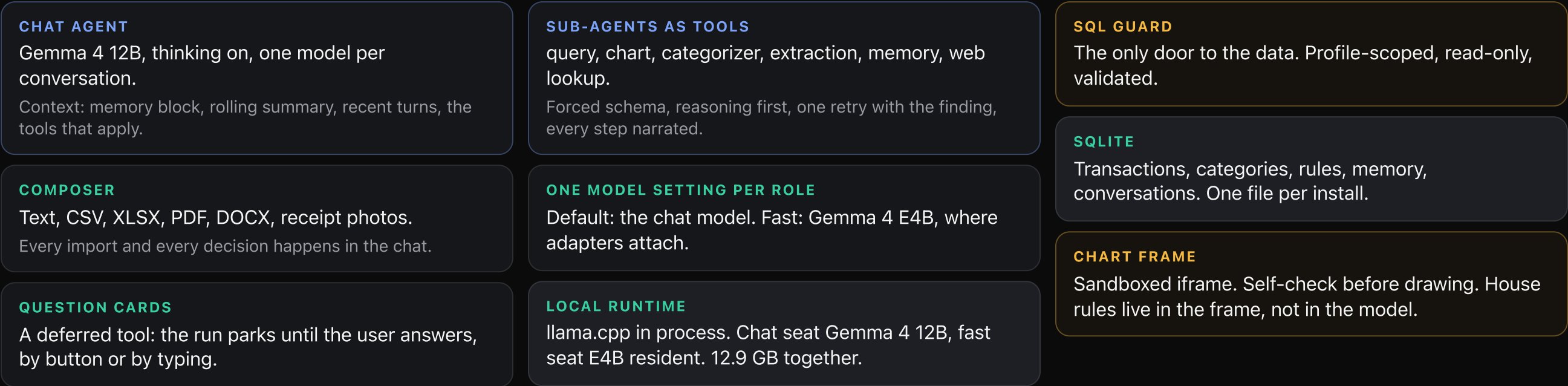
THE ONE RULE

Numbers come from executed queries, never from the model



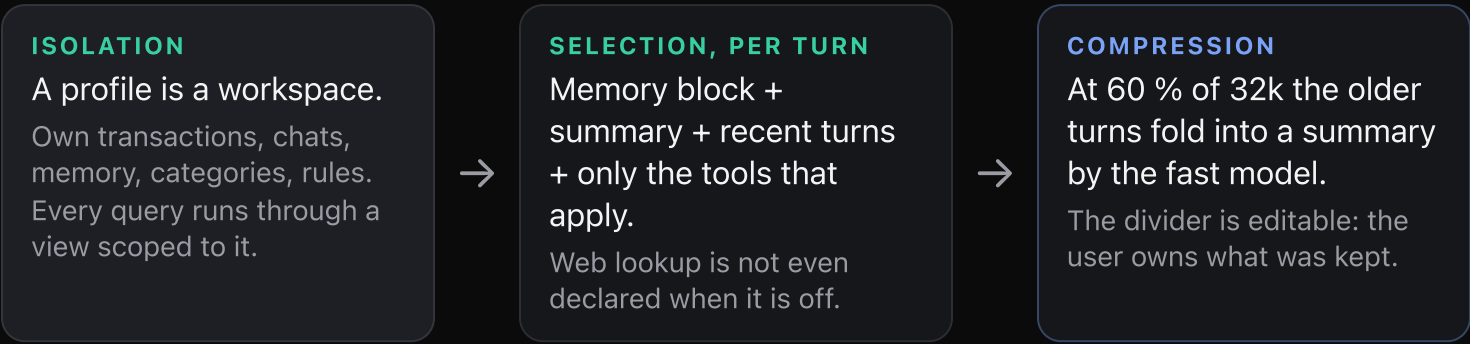
- **The model writes the question.** The database writes the numbers.
- **Charts take the same path:** generated code, self-checked on the real rows, drawn in a sandbox.
- **Measured:** the check pass alone lifts figure match from 69 to 79 percent on the same model.

One chat agent, six sub-agents as tools, a guard, a sandbox



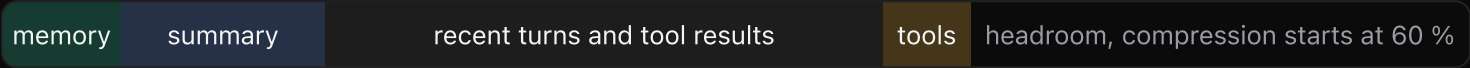
Intelligent context management

isolation, selection, compression



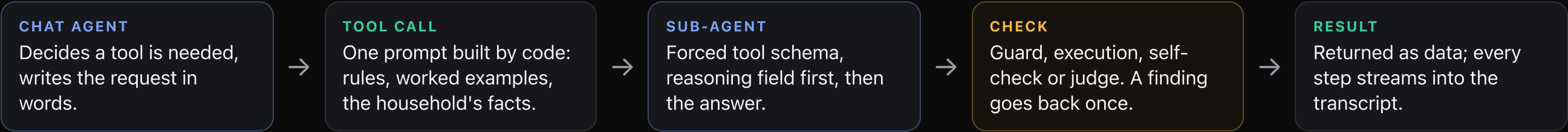
- **Memory is distilled** after every turn: at most two durable facts, never figures or dates unless the user states a rule.
- **Selection is by need**, not by recency alone.
- **Compression is visible** and reversible in the transcript.

What one turn's 32k budget holds



Sub-agent deployment

an LLM calls another LLM for one task

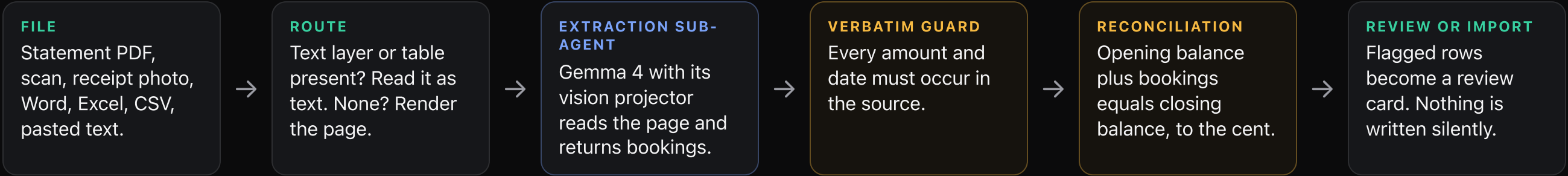


SUB-AGENT	GETS	RETURNS	CHECKED BY
query	request, schema, rules, examples	reasoning + SQL	guard, execution, check pass
chart	request, rows brief	plan, then chart code	self-check on real rows, frame
categorizer	batch of unknown bookings	category + confidence per booking	threshold, Question cards, rules
extraction	a page as text or image	bookings	verbatim and reconciliation guards
memory	the finished turn	at most two facts	no figures, no dates
web lookup	a scrubbed merchant token	search, fetch or finish	floors, verbatim quote, cap

Not claimed: the extra credit for a controller that continues without waiting. The turn survives a closed tab as a server task, but the chat agent waits for its tool.

Multimodal ingestion

PDF, images, text, CSV, XLSX, DOCX



- **Inherently multimodal:** the same local models read images and text.
- **The model points, code verifies.** It may not invent a digit.
- **Every duplicate is asked about** before anything lands.
- **19 of 19** receipt totals right on real and web receipts.
- **Excel** feeds the CSV mapping engine; **Word** feeds the text path.
- **Audio** is out of scope, and we say so.

Self-controlled web search

find a page, read it, quote it

SCRUBBER

Booking text becomes a merchant token. A person's name is refused. Nothing else ever leaves.



DECIDE

Search, fetch or finish, with reasoning. Up to 4 searches and 3 page reads.



FLOORS IN CODE

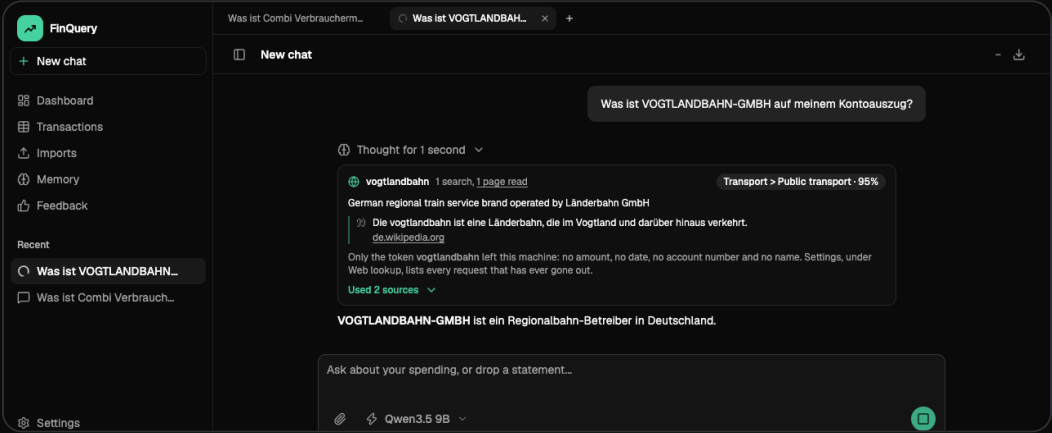
At least one search. A page read when a hit is the merchant's own site or Wikipedia. A failed backend is retried free.



FINISH

A verbatim quote that must occur in what was read. Confidence capped when no source names the merchant.

- Every request is journaled before it is sent and shown in Settings.
- Three callers, one loop: the chat tool, the categorizer at import, receipt headers.
- **Cached per profile:** the second question about a merchant leaves nothing.
- **Measured:** lookups without a search 9 to 0, pages read 0 to 12 after the floors went in.



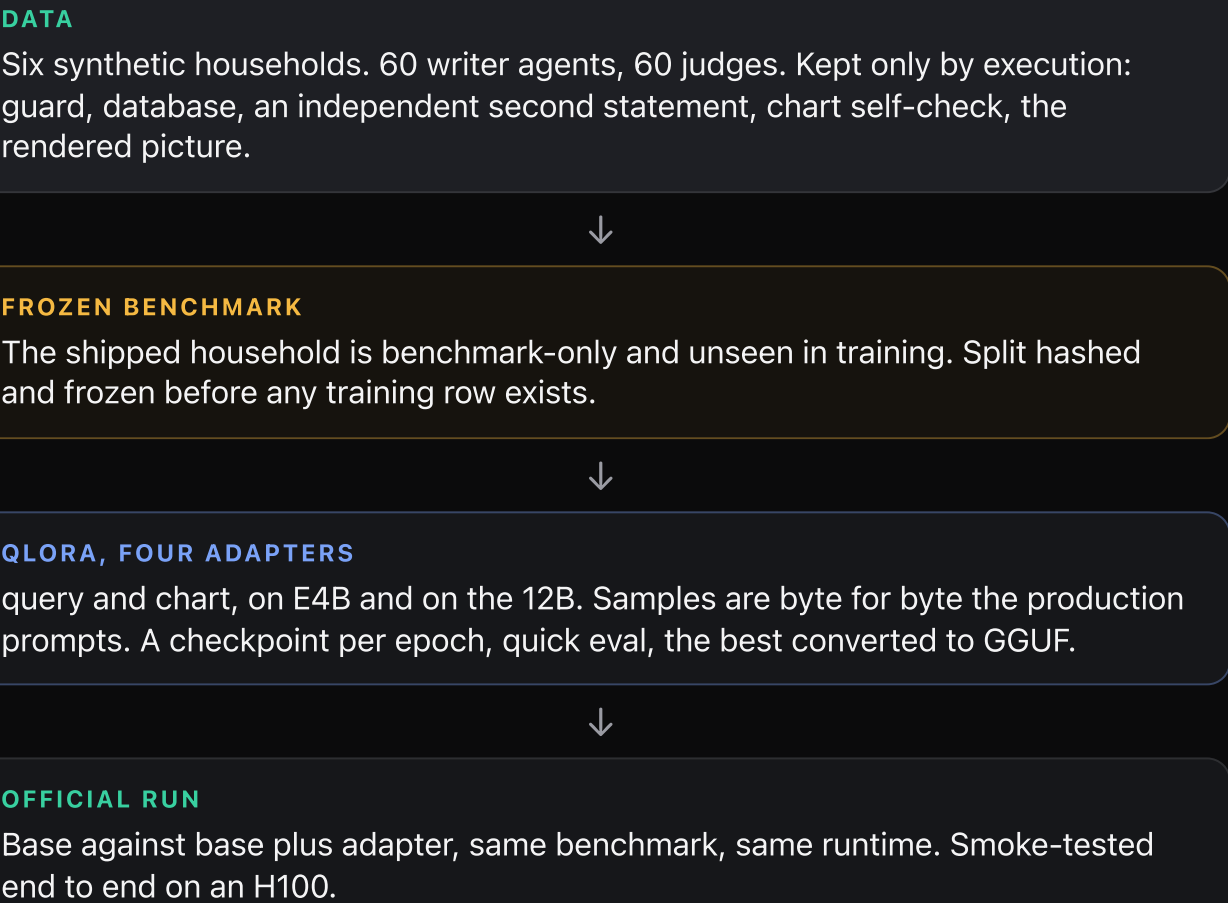
Which local models, and how the adapters are trained

MODEL, Q4_K_M	SQL	CHARTS	END TO END	TOK/S
Gemma 4 E4B	66 %	45 %	53 %	47
Qwen3.5 9B	70 %	42 %	73 %	30
Gemma 4 12B	87 %	81 %	77 %	22

Figure match on 152 SQL and 73 chart cases, held-out third never trained on. Same GGUFs and wire formats as the laptop.

- **Rule written before the numbers:** accuracy first, speed within five points, the pair must fit in 13.5 GB.
- **Result:** Gemma 4 12B for chat and sub-agents, E4B resident as the fast seat and adapter target.

The fine-tuning loop **in progress**



WHAT WE OWN

Decisions we made, and the ones we reversed on evidence

KEPT

Numbers only from executed SQL. Sub-agents as forced tool calls, reasoning first. Every import and decision in the chat. Training equals production: a sample is the exact prompt the sub-agent sends.

NOT CLAIMED, ON PURPOSE

Sub-agent extra credit. Python code execution (our sandbox runs JavaScript charts). Audio ingestion.

docs/explainers: one chapter per required bullet and per elective, with sources. docs/adr: 13 decisions. bench: the benchmark and every run. training: the data harness and the cluster loop.

REVERSED

Qwen3.5 9B chosen on memory, replaced by Gemma 4 12B after the cluster benchmark. Preference optimisation built, measured and removed for multimodal ingestion. "No dashboard" became a dashboard of guarded queries.

NEXT

Adapter numbers against vanilla. The video. A second data round if the learning curve asks for it.